

Neural Network Models

Emanuela Raffinetti
Financial Data Science

Department of Economics and Management
University of Pavia

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Outline

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- 4 Predictions

Background

- Neural networks have been developed in the field of machine learning, imitating the neurophysiology of the human brain.
- Neural networks are information processing systems, composed of a large number of highly interconnected elements (neurons), working in union to solve specific problems.
- There are two main kinds of neural networks: supervised and non-supervised (self-organising).
- Here we consider supervised neural networks.

Architecture of a neural network - I

- The most common structure of a supervised network is composed of a graph, where the nodes (*neurons*) are placed on more levels (*layers*) and are interconnected from a layer to the other in a single direction.

Architecture of a neural network - II

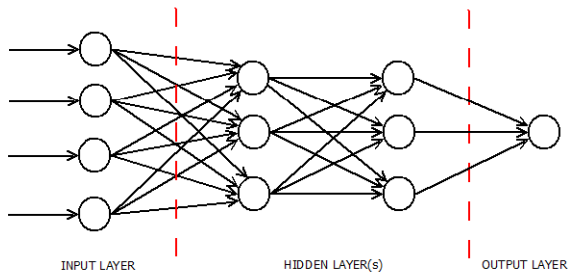


Figure: Structure of a supervised neural network

Architecture of a neural network - III

- The neurons of a neural network are organized in three types of layers: (i) input, (ii) output, (iii) hidden.
- (i) The **input** layer receives information from the external environment; each neuron in it usually corresponds to an explanatory variable.
- (ii) The **output** layer furnishes the final results, which are sent by the network to the outside of the system. Each of its neurons corresponds to a response variable.
- (iii) The **hidden** layers contain intermediate computational neurons, whose role is to increment the model fit.

How does a neural network work? - I

- A generic neuron j , with a threshold θ_j , receives n input signals $x = [x_1, x_2, x_3, \dots, x_n]$ from the units it is connected to in the previous layer.
- Each signal has an importance weight:
 $w_j = [w_{1j}, w_{2j}, w_{3j}, \dots, w_{nj}]$
- The same neuron elaborates the input signals, according to their importance weights and the threshold value, through a function called *combination function*, that produces a value called *potential* or *net input*.
- The potential of a neuron j is defined by the following linear combination:

$$P_j = \sum_{i=1}^n (x_i w_{ij} - \theta_j)$$

How does a neural network work? - II

- Consider now the way according to which every neuron sends signals in output.
- The output of the j -th neuron, y_j , derives from the application of a function, called *activation function*, to the potential P_j :

$$y_j = f(x, w_j) = f(P_j) = f\left(\sum_{i=0}^n x_i w_{ij}\right)$$

How does a neural network work? - III

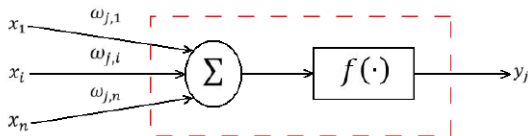


Figure: Signal elaboration in a neuron

Single-layer perceptrons

- Neural networks with a single set of weights are known as **single layer perceptrons**.
- They are constituted by n input units ($x_1, x_2, x_3, \dots, x_n$) connected to a layer of p output units ($y_1, y_2, y_3, \dots, y_p$) through a system of weights that can be represented in matrix form, as follows:

$$\begin{array}{ccccc} w_{11} & \dots & w_{1j} & \dots & w_{1p} \\ \dots & \dots & \dots & \dots & \dots \\ w_{i1} & \dots & w_{ij} & \dots & w_{ip} \\ \dots & \dots & \dots & \dots & \dots \\ w_{n1} & \dots & w_{nj} & \dots & w_{np} \end{array}$$

w_{ij} represents the weight of the connection between the i -th neuron of the input layer and the j -th neuron of the output layer

Multi-layer perceptrons

- Neural networks with more than one set of weights are called **multi-layer perceptrons**.
- Consider the simplest case, with n neurons in the input layer, h in the (only) hidden layer and p in the output layer.
- The network can be represented by a double system of weights: the weights w_{ik} ($i = 1, \dots, n; k = 1, \dots, h$) that connect the nodes of the input layer with the hidden ones and the weights z_{kj} ($k = 1, \dots, h; j = 1, \dots, p$) that connect the nodes of the hidden layer with the output nodes.

Multi-layer perceptrons

- The output of a generic neuron j of the output layer is therefore:

$$y_j = g \left(\sum_k l_k z_{kj} \right) = g \left(\sum_k z_{kj} f \left(\sum_i x_i w_{ik} \right) \right)$$

- A deep learning network is a neural network with many layers, described by many compound functions.

Estimates and Predictions

- The weight parameters of a neural networks are typically unknown, and should be estimated from the data.
- However, the final aim is not to estimate some parameters, but to predict future values.
- The performance of a neural network can be measured by splitting the data into a training and a validation set.
- The best architecture will then be chosen by maximising the obtained performance measure.

Training of a neural network - I

- Once decided the structure of a neural network, it is necessary to *train* it.
- This process is based on a training set which is composed of an input vector and a target vector, and is called *supervised learning*.

Training of a neural network - II

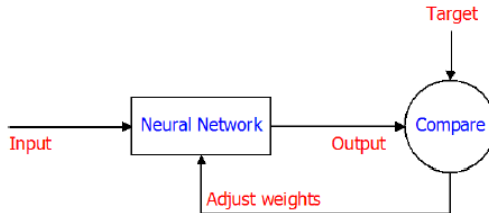


Figure: Supervised learning in a neural network